Non-Teaching Hours and Teacher Stress in Singapore: Non-Linear Evidence from TALIS 2024

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Abstract

Singapore's Parliament has urged MOE to set measurable targets for non-teaching hour reduction, amid rising stress among lower-secondary teachers. Meaningful targets require task-specific evidence on where reductions matter most, yet no Singapore study has examined the non-linear workload-stress relationship at the task level. Using TALIS 2024 Singapore data (N = 1,706) and survey-weighted regressions with cluster-robust errors, this study finds that marking carries the strongest stress association: each additional hour predicts a 0.065-unit rise on the workload stress scale. Crucially, the steepest stress gradient lies at low hours rather than high ones. Below a 3-hour weekly threshold, each marking hour is associated with over thirteen times the stress of an hour above it, and admin hours follow the same pattern. These findings suggest that hours-based targets focused solely on the most burdened teachers may miss where stress rises fastest, though causality cannot be established from this cross-sectional design.

Keywords: Workload Stress, TALIS 2024, Singapore, non-teaching hours, quadratic survey-weighted regression, non-linear workload-stress association

1 Introduction

Singapore teachers work among the longest hours of any OECD education system, averaging 47.3 hours per week (third highest), yet spend only 17.7 of those hours in actual classroom instruction, below the OECD average of 22.7 hours (OECD, 2025d). The remaining hours are absorbed by a wide range of non-teaching duties: lesson planning, marking and correcting students' work, teamwork with colleagues, general administrative tasks, school management, parent communication, student counselling, professional management, co-curricular activity (CCA) supervision and other miscellaneous tasks. TALIS 2024 further shows that 27% of Singapore teachers report experiencing stress "a lot" at work, up from 23% in 2018, a rise of 4% over 6 years (OECD, 2025d).

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Singapore teachers most commonly cite administrative work (53%), marking (49%), and being held accountable for student outcomes (45%) as sources of stress (OECD, 2025d). In November 2025, Members of Parliament raised this issue directly in Parliament, calling on MOE to set specific, measurable targets for non-teaching hour reduction (MOE, 2025a). However, meaningful targets require task-specific evidence on which non-teaching duties drive stress and at what margins. This study estimates how hours spent on each non-teaching task are associated with workload stress among Singapore lower-secondary teachers. Using TALIS 2024 Singapore data (N = 1,706), I estimate survey-weighted linear and quadratic regressions with school-level cluster-robust standard errors to test whether the stress-workload association differs across tasks and whether it is non-linear. Marking has the strongest positive association with workload stress, with each additional hour associated with a 0.065 unit increase in the workload stress scale. The quadratic specification implies turning points at 18.3 and 12.4 hours for marking and admin respectively, but both sit above the 95th percentile of observed hours and are better read as extrapolations than as actionable thresholds. A piecewise specification instead finds the steepest stress gradient at low hours, with the marking and admin slopes falling roughly 13.5-fold and 12.7-fold above their respective thresholds. Teamwork, counselling, and school management hours are negatively associated with stress. CCA hours show no significant independent association once other tasks are held constant. These findings are robust to a piecewise linear specification and an ordered probit estimator. This paper contributes to the economics of education literature in three ways. First, it provides the first Singapore-specific, task-level evidence on the non-linear workload-stress association using TALIS 2024. Second, it examines CCA and parent communication as predictors of stress, two tasks unstudied in the Singapore literature despite growing since 2018. Third, its threshold estimates directly inform the measurable targets Parliament requested in November 2025. The remainder of the paper is structured as follows. Section 2 covers Singapore's institutional context and the theoretical framework. Section 3 describes the data source, sample construction, variable definitions, and descriptive statistics. Section 4 sets out the model specifications, empirical strategy, and robustness checks. Section 5 presents and interprets the empirical results. Section 6 concludes and discusses policy implications.

2 Background

2.1 Institutional Context

Singapore's MOE has undertaken successive workload reduction efforts since at least 2018, including the Teacher Work Management Framework (TWMF), centralised administrative support, and AI-assisted marking tools (MOE, 2019; MOE, 2025b). Despite these interventions, administrative hours have remained at approximately 4 hours per week, and overall working hours have stayed stable at around 47 hours per week since 2018 (MOE, 2025a). CCA supervision and parent communication are recognised components of teacher workload in Singapore secondary schools, features

not examined in the five Anglophone systems studied by Jerrim and Sims (2021), reflecting Singapore's distinctive workload structure. The hours spent on both these tasks have increased since 2018 (MOE, 2025b). Earlier Singapore evidence confirms that teachers report the highest work stress across professional groups (Chan et al., 2000), and that marking carries an emotional labour burden among Singapore English teachers (Loh & Liew, 2016). The persistence of rising stress despite repeated policy interventions points to the need for task-specific evidence on where workload reductions would matter most.

2.2 Theoretical Framework

This study draws on the Job Demands-Resources (JD-R) model (Bakker & Demerouti, 2007) as its theoretical foundation. The model organises work characteristics into two broad categories. Demands refer to aspects of work that call for sustained physical or cognitive effort. Resources, by contrast, are conditions that help workers meet those demands, achieve their goals, and grow professionally. The JD-R model identifies two opposing processes through which these categories shape employee wellbeing. When demands are high and prolonged, workers' psychological reserves become depleted, increasing the risk of burnout and, over time, poorer health outcomes. When resources are sufficient, workers remain motivated and invested in their roles, leading to greater engagement and organisational commitment. Hakanen et al. (2006) tested this model among 2,038 Finnish teachers and confirmed both processes. Excessive workload, disruptive student behaviour, and poor working conditions predicted health decline through burnout, while autonomy, social support, and a supportive school climate strengthened engagement and commitment. In the context of this study, non-teaching tasks represent job demands. Time spent on marking, administrative work, CCA supervision, and parent communication all require sustained effort and draw on finite cognitive and emotional capacity. This provides a theoretical basis for expecting a positive association between non-teaching hours and workload stress, with the strength of that association potentially varying by task type.

3 Data

3.1 Data Overview

This study uses the OECD's 2024 Teaching and Learning International Survey, filtered to Singapore country (OECD, 2025e). The Singapore TALIS 2024 dataset contains 3,355 lower-secondary teacher observations, drawn from a near-census of all 145 public secondary schools and 10 randomly selected private secondary schools (MOE, 2025b). The respondents filled out the TALIS survey between April to August 2024. Sampling weights (TCHWGT) are applied throughout to produce nationally representative estimates. Cluster-robust standard errors are applied at the school level (IDSCHOOL) to account for the non-independence of teachers within the same school.

3.2 Sample Construction

TALIS 2024 Singapore used a rotation design where the task-hour questions (TT4G16A-TT4G16J), workload stress measure (T4WLOADT), and employment status variables were only given to about two-thirds of teachers. The remaining 1,126 teachers did not receive this section by design. OECD (2025e) confirms that form assignment was random and unrelated to any teacher or school characteristics, supporting the classification of this missingness as Missing Completely At Random (MCAR). To verify the MCAR assumption empirically, a logistic regression was conducted on 3,281 teachers in which rotation-form receipt (1 = received, 0 = not received) was regressed on five available teacher-level covariates: gender, teaching qualification, collaborative culture, peer support, and multi-school status. Results showed that four of the five covariates had no significant relationship with whether a teacher received the rotation form: gender ($p = 0.156$), qualification ($p = 0.379$), collaborative culture ($p = 0.544$), and peer support ($p = 0.426$). This is consistent with MCAR. The only exception was multi-school status. Teachers working across more than one school were significantly less likely to have received the rotation form ($\beta = -0.813, p < 0.001$), which caused the overall joint test to be significant ($LR\chi^2 = 23.92, df = 5, p < 0.001$). However, this finding should be interpreted with caution for two reasons. First, multi-school teachers represent only 4.1% of the Singapore sample. Second, the model's Pseudo R^2 was just 0.006, meaning all five covariates together explained less than 1% of the variation in form receipt. The overall pattern of missingness therefore remains consistent with a random rotation design. This minor departure from strict MCAR is acknowledged as a limitation but is not considered as a threat to the validity of the findings. It is important to distinguish rotation-induced missingness from item-level non-response within the rotation form. Among teachers who received the relevant questionnaire block, 461 did not provide complete responses across all analysis variables. This item-level non-response is not guaranteed to be MCAR. Listwise deletion is applied across all analysis variables, removing these 461 teachers alongside the 1,126 rotation non-recipients, retaining 1,768 teachers with complete data.

Teachers reporting total weekly non-teaching hours exceeding 84 are then excluded as implausibly high reporters; this removes 62 teachers (3.5%), yielding the final analysis sample of 1,706. All ten non-teaching task variables (TT4G16A through TT4G16J) are retained in the analysis, and they were administered on the same rotated questionnaire form. TT4G16J (other tasks), which has a mean of 8.1 hours per week, is included as a residual control to prevent omitted-variable bias.

3.3 Variable Definitions

3.3.1 Dependent variable

The outcome variable is the Workload Stress Scale (T4WLOADT), a four-item OECD-constructed composite scale. Respondents indicate the extent to which each of four statements applies to them,

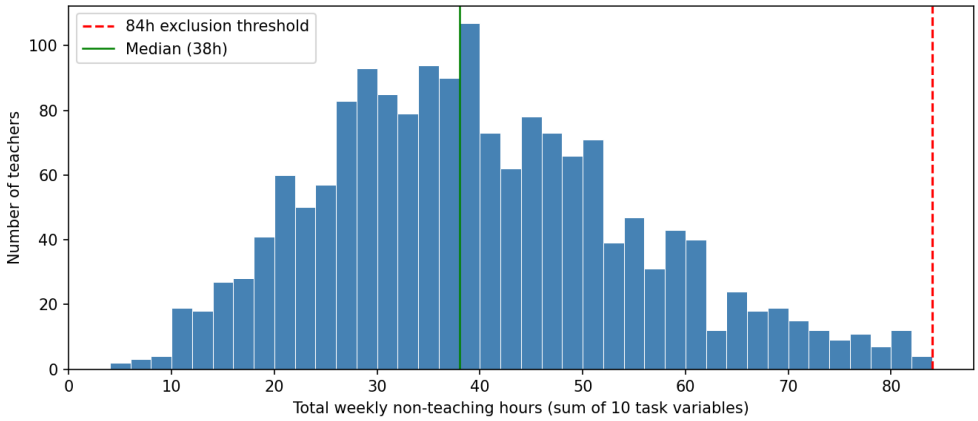


Figure 1: Distribution of total weekly non-teaching hours.
Notes: Full sample (N = 1,706) after 84-hour exclusion, weighted by TCHWGT. Total hours computed as the sum of 10 non-teaching task variables (TT4G16A–TT4G16J). Red dashed line marks the 84-hour exclusion threshold. Green line marks the median (38h).

across response options of not at all, to some extent, quite a bit, and a lot. The TALIS 2024 version is constructed using multi-group confirmatory factor analysis and standardised to a mean of 10 and standard deviation of 2 across all participating countries (OECD, 2025c). The analogous scale in TALIS 2018 reported an Omega reliability above 0.85 across participating countries (Jerrim & Sims, 2021), and the TALIS 2024 version provides a more reliably constructed measure of workload stress than its 2018 counterpart. Singapore's weighted mean on T4WLOADT is 9.77 (SD = 1.74), marginally below the international mean of 10. Regression coefficients will be reported in: (i) raw units of the T4WLOADT scale; (ii) fractions of the Singapore-specific standard deviation of 1.74; and (iii) fractions of the international standard deviation (SD = 2). Concerns about cross-country comparability, raised by Jerrim and Sims (2021), are not relevant here, as this study uses data from Singapore only. Within a single system, the TALIS stress scale captures the same construct across respondents. TALIS 2024 scales are constructed using common international model parameters, supporting their internal consistency and validity (OECD, 2025c).

3.3.2 Task-hour variables

All task-hour variables are drawn from the TALIS 2024 Singapore-adapted questionnaire (OECD, 2024), in which teachers were asked: "Approximately how many 60-minute hours did you spend on the following tasks during your most recent complete calendar week, in your job at this school?" The ten non-teaching task variables used are: (i) individual lesson planning or lesson preparation (TT4G16A); (ii) teamwork and dialogue with colleagues (TT4G16B); (iii) marking/correcting student work (TT4G16C); (iv) counselling student (TT4G16D); (v) participation in school management (TT4G16E); (vi) general administrative work (TT4G16F); (vii) professional learning activities (TT4G16G); (viii) communication and co-operation with parents or guardians (TT4G16H); (ix) en-

gaging in extracurricular activities (TT4G16I); and (x) other work tasks (TT4G16J). Singapore administered this question in open-text format, with teachers entering exact integer hours.

3.3.3 Control variables

The primary specification includes the following control variables, selected based on theoretical relevance and availability in the Singapore TALIS 2024 public-use file. Demographic controls are included because teacher-level characteristics may be correlated with both task-hour allocation and stress, creating omitted variable bias if excluded. Such control variables are gender, employment status, contract type, teacher qualification, number of other schools they are working at, teaching experience at the current school and total teaching experience. Female teachers report higher stress on average (OECD, 2025d); full-time teachers carry heavier non-teaching obligations; and permanent teachers may face greater institutional pressure to perform non-teaching duties. Total teaching experience (T4TYEXPTT) and experience at the current school (T4TYEXPTS) are included as ordinal grouped variables ($\leq$5 years, 6-10 years, 11-20 years, above 20 years) because more experienced teachers may manage task demands more efficiently, reducing stress independently of hours. Two job resource controls are included as per the JD-R model, which holds that resources buffer the negative effect of demands on stress. Collaborative school culture (TT4G64D) and peer support (TT4G64H) are both negatively associated with teacher stress at work after controlling for teacher and school characteristics (OECD, 2020). Several demographic controls remain unavailable in the Singapore TALIS 2024 public-use file. Teacher age (TT4G02) is suppressed from the public-use file internationally across all participating systems. The grouped age variable (T4TAGEGR), educational attainment (T4THEDAT), and subject specialism (TT4G48A) are suppressed specifically for Singapore (OECD, 2025e).

3.3.4 Descriptive Statistics

Table 1 presents weighted descriptive statistics for all non-teaching task-hour variables and the workload stress outcome for the analysis sample (N = 1,706), after application of the 84-hour exclusion. 'Other work tasks' has the highest mean at 8.1 hours per week (SD = 6.3h), followed by planning at 7.6h (SD = 5.2h) and marking at 5.9h (SD = 4.5h). Teamwork with colleagues averages 3.4h (SD = 2.4h) and general admin averages 3.5h (SD = 3.0h). Among the tasks specific to Singapore, CCA averages 2.9h (SD = 2.5h) and parent communication 1.4h (SD = 1.3h). Notably, 44% of respondents reported 0 hours for participation in school management, 21.9% reported 0 CCA hours and 15% report 0 parent communication hours.

Table 1: Weighted descriptive statistics (non-teaching task-hour variables and workload stress, Singapore TALIS 2024)

Variable	Mean	SD	p25	p50	p75	p95	Zeros (%)	N
A. Non-teaching task-hour variables (hours per week)								
Planning	7.6	5.2	4.0	6.0	10.0	18.0	0.1	1706
Teamwork with colleagues	3.4	2.4	2.0	3.0	5.0	8.0	1.9	1706
Marking / correcting	5.9	4.5	3.0	5.0	8.0	15.0	3.9	1706
Student counselling	2.4	2.3	1.0	2.0	3.0	5.1	5.1	1706
School management	1.7	2.6	0.0	1.0	3.0	6.0	44.0	1706
General admin	3.5	3.0	2.0	3.0	5.0	10.0	2.2	1706
Professional development	1.8	2.2	1.0	1.0	2.0	5.0	18.1	1706
Parent communication	1.4	1.3	1.0	1.0	2.0	3.0	15.0	1706
CCA	2.9	2.5	1.0	3.0	4.0	7.0	21.9	1706
Other tasks	8.1	6.3	4.0	7.0	11.0	20.0	2.3	1706
B. Outcome variable								
Workload stress	9.77	1.74	8.8	9.7	10.9	12.5	—	1706

Notes: Analysis sample (N = 1,706), weighted by TCHWGT, after 84-hour total non-teaching hours exclusion. Mean, SD, and percentiles are survey-weighted; hour variables recorded in whole hours. Zeros (%) is the unweighted share of valid observations equal to zero; — not applicable.

Figure 2 presents the weighted correlations ranked by magnitude. Marking shows the strongest association with workload stress ($r = 0.19$), followed by other tasks ($r = 0.13$) and planning ($r = 0.10$). With the exception of school management ($r = -0.13$), no remaining task variable exceeds an absolute correlation of 0.06 with workload stress, suggesting that time spent on CCA, administration, professional development, teamwork, counselling, and parent communication carries limited independent stress signal in bivariate analysis. The full weighted pairwise correlation heatmap for all analysis variables is depicted in Figure A4. The median total weekly non-teaching hours across the ten task variables is $38\,h$ in the analysis sample.

The bivariate association between marking hours and workload stress is shown in Figure 3. The positive correlation (r = 0.19) is the strongest among all ten non-teaching task variables, consistent with prior evidence that marking is strongly associated with teacher stress across five Anglophone systems. Figure A1 shows the distribution of workload stress scale scores for the analysis sample (N = 1,706). The Singapore weighted mean stress of 9.77 is marginally below the international standardised mean of 10. The pattern for CCA hours is non-monotonic, as shown in Figure 4, consistent with a concave association and motivating the quadratic specification in Model 2. Figure 5 shows a similar pattern for general administration hours, with stress rising between Q1 and Q2 before plateauing across higher quartiles.

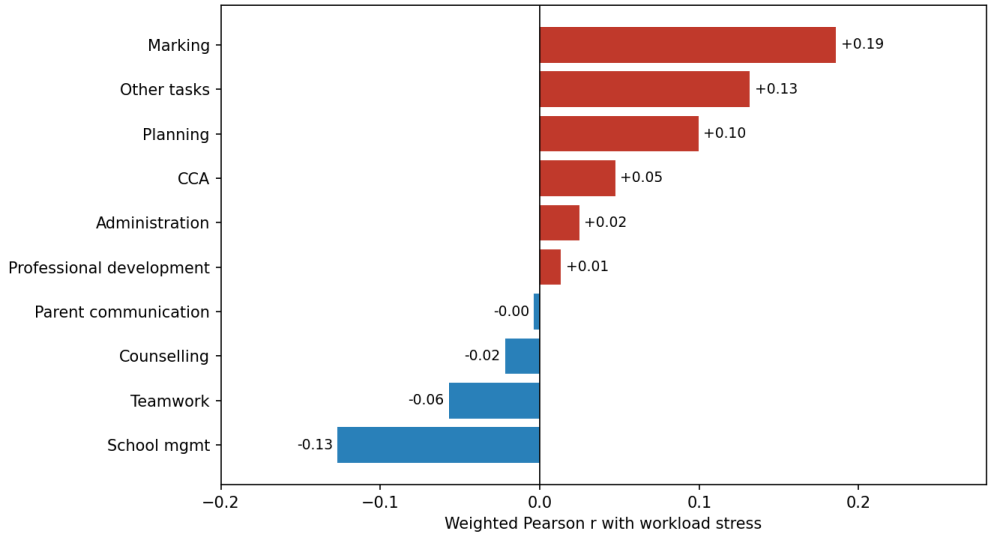


Figure 2: Survey-weighted Pearson correlations between each non-teaching task hours variable and workload stress (T4WLOADT), Singapore TALIS 2024

Notes: Analysis sample (N = 1,706), weighted by TCHWGT.

3.4 Data Limitations

This study has several data limitations. First, all task-hour variables are self-reported and subject to recall error. Second, several demographic controls remain unavailable in the Singapore TALIS 2024 public-use file i.e., teacher age (TT4G02) is suppressed internationally, while the grouped age variable (T4TAGEGR), educational attainment (T4THEDAT), and subject specialism (TT4G48A) are suppressed specifically for Singapore (OECD, 2025e). Third, item-level non-response among rotation-form recipients is not guaranteed to be MCAR, and listwise deletion may introduce selection bias if non-response is systematically related to unobserved teacher characteristics. Fourth, the OECD reports Singapore teachers average 47.3 working hours per week (OECD, 2025d), implying roughly 29.6 non-teaching hours once classroom teaching time is excluded. However, the sum of the ten task-hour variables used in this study (Table 1) averages 38.7 hours per week (median 38h), roughly nine hours higher. Three explanations are plausible and not mutually exclusive. First, the ten task categories are not mutually exclusive by design, a single activity can plausibly belong to more than one category. A teacher who plans a lesson as part of a professional development programme, for instance, may report the same hours under both planning and professional development, inflating the total. Second, "other work tasks" (TT4G16J) is the single largest category at 8.1 hours per week. Unlike the other nine categories, it carries no definition of what qualifies, giving respondents wide latitude in what they report. Some of these reported hours likely reflect overestimation rather than a genuinely distinct activity, which would inflate the overall total. Third, the OECD's 47.3-hour reported total may itself derive from a different survey item, specifically a single total-hours question, rather than from summing the same ten task items used here. If so, the two measures are not directly comparable by construction, and the gap may reflect a measurement

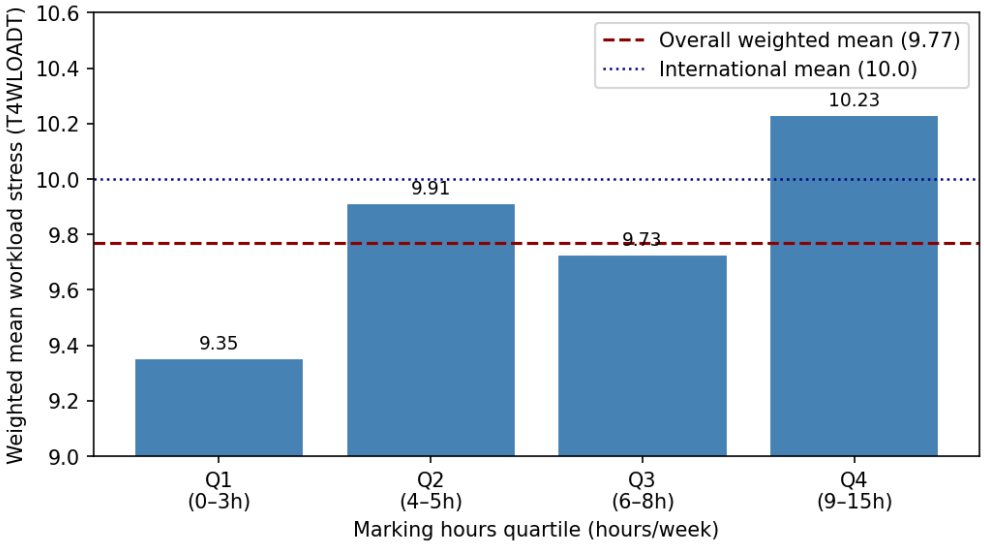


Figure 3: Weighted mean workload stress by marking hours quartile, Singapore TALIS 2024
Notes: Analysis sample (N = 1,706), weighted by TCHWGT. Red dashed line is the overall weighted mean (9.77). International mean = 10.0.

difference rather than a genuine discrepancy in workload. This study relies on the ten task-hour items throughout, both because they are the only task-specific measures available and because the regression results depend on relative, not absolute, hours across tasks. The discrepancy is noted here as a limitation on how accurately the reported hour levels reflect true workloads, though it does not necessarily affect the estimated associations between hours and stress. Fifth, the 84-hour exclusion threshold (Section 3.2) removes 62 teachers (3.5% of the pre-exclusion sample) reporting implausibly high total non-teaching hours. To assess sensitivity, Model 2 is re-estimated without this exclusion (N = 1,768), which reduces the marking and admin coefficients by approximately 31% and 35% respectively. A stricter 70-hour cutoff (N = 1,646) leaves both within 9% of the main specification. Sign and significance are unaffected in either case.

4 Empirical Strategy

4.1 Selection of Tasks and Identification

Conditional associations between task-level hours and workload stress are estimated. Following Angrist and Pischke (2009), the WLS coefficients estimate the average stress difference between teachers who differ in hours on a given task, holding observable characteristics constant. Prior evidence motivates both the task selection and the non-linear specification. Jerrim and Sims (2021) find that across five Anglophone systems, marking is the task most strongly associated with workload stress, with a conditional effect of 0.059 standard deviations per additional hour, while management and administrative hours show little association. The workload-stress relationship may

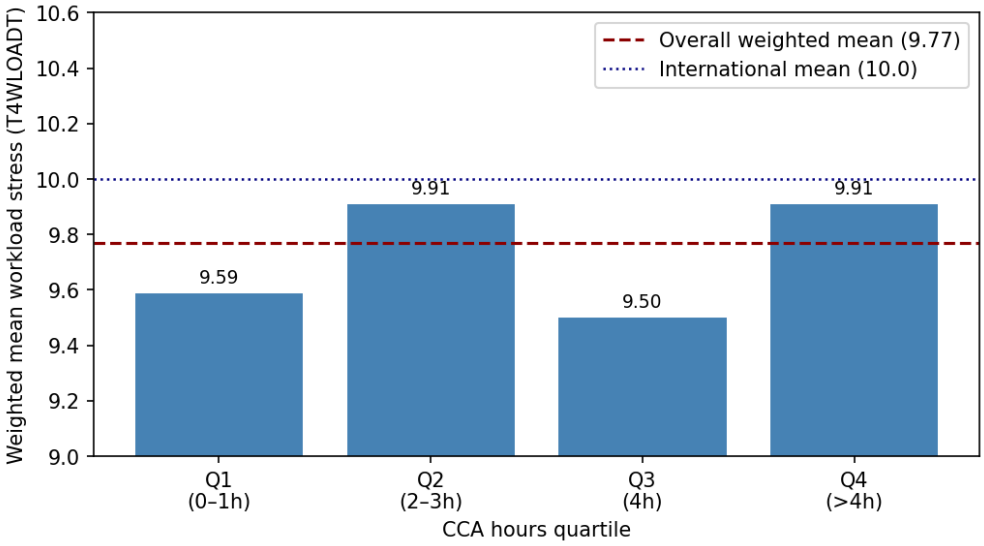


Figure 4: Weighted mean workload stress by CCA hours quartile, Singapore TALIS 2024

Notes: Analysis sample (N = 1,706), weighted by TCHWGT. Red dashed line is the overall weighted mean (9.77). International mean = 10.0.

also be non-linear, with teacher wellbeing declining once working hours exceed 55 hours per week in some countries. Whether such thresholds exist in Singapore, and where stress accelerates, remains an open empirical question. Three tasks are selected for the quadratic non-linearity test in Model 2. Marking is selected on empirical grounds, it has the strongest bivariate association with stress in this sample (r = 0.19). Administrative work is selected on policy grounds, despite successive MOE interventions, administrative hours have remained at approximately 4 hours per week since 2018 (MOE, 2025a), and Figure 5 shows stress rising between Q1 and Q2 before plateauing, consistent with a concave association. CCA is selected on institutional grounds, CCA supervision is a recognised component of teacher workload in Singapore secondary schools, hours have increased since 2018 (MOE, 2025b), and Figure 4 shows a non-monotonic stress pattern. Japanese teachers spend around 8 hours weekly on extracurricular duties, roughly four times the TALIS average of 2 hours (OECD, 2015). Additionally, parent communication is included as a standalone predictor; neither CCA nor parent communication has been examined in the peer-reviewed quantitative literature for Singapore, and the OECD's own regression analysis found parent communication is one of three task types significantly associated with decreased teacher wellbeing across all TALIS systems (OECD, 2025b).

4.2 Model Specifications

All models are estimated through weighted least squares (TCHWGT as the weight variable), with the Workload Stress Scale (T4WLOADT) as the dependent variable throughout and cluster-robust standard errors at the school level (IDSCHOOL). Before estimating task-specific associations,

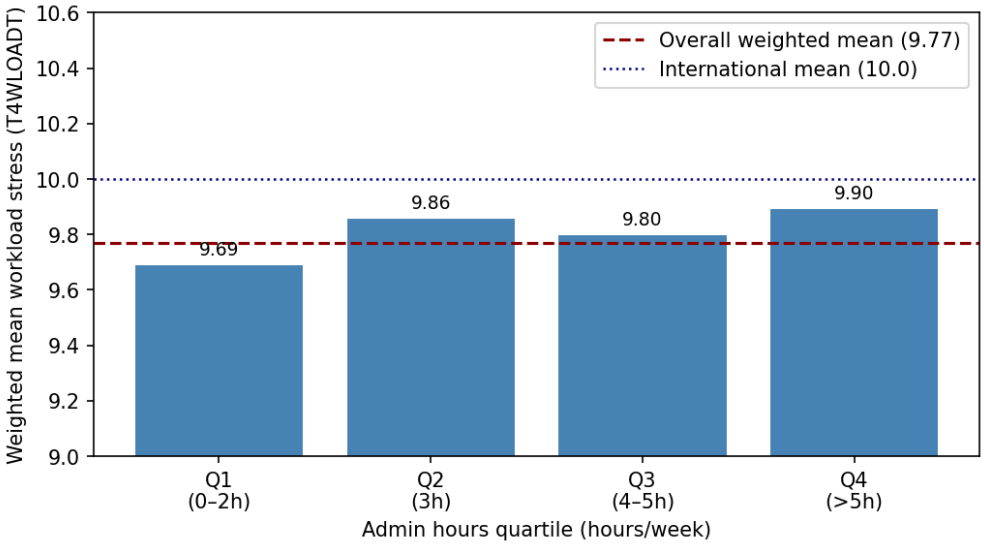


Figure 5: Weighted mean workload stress by general administration hours quartile, Singapore TALIS 2024

Notes: Analysis sample (N = 1,706), weighted by TCHWGT. Red dashed line is the overall weighted mean (9.77). International mean = 10.0.

Model 0 examines whether teaching and non-teaching hours are associated with workload stress differently. If non-teaching hours drive stress more than teaching hours, this motivates the task-disaggregated analysis in Models 1 and 2. Model 0 takes the form:

$$Stress_i = \beta_0 + \beta_1 TeachingHours_i + \beta_2 TotalNonTeachingHours_i + \gamma X_i + \varepsilon_i \quad (0)$$

where $TeachingHours_i$ is self-reported classroom teaching hours, $TotalNonTeachingHours_i$ is the sum of all ten non-teaching task variables (TT4G16A to TT4G16J), X_i is the vector of controls defined in Section 3.3.3, and ε_i is the error term.

Model 1 estimates which tasks are most strongly associated with stress under a linear specification:

$$\begin{aligned} Stress_i = \beta_0 &+ \beta_1 Planning_i + \beta_2 Teamwork_i + \beta_3 Marking_i + \beta_4 Admin_i + \beta_5 Parents_i \\ &+ \beta_6 Counselling_i + \beta_7 CCA_i + \beta_8 Professional Development_i + \beta_9 School Management_i \\ &+ \beta_{10} Other Tasks_i + \gamma X_i + \varepsilon_i \quad (1) \end{aligned}$$

where X_i denotes the vector of control variables and ε_i the error term.

Model 2 extends the linear baseline by adding quadratic terms for marking, administrative work,

and CCA hours simultaneously. The quadratic specification takes the form:

$$\begin{aligned} Stress_i = \beta_0 &+ \beta_1 Planning_i + \beta_2 Teamwork_i + \beta_3 Marking_i + \beta_4 Marking_i^2 + \beta_5 Admin_i + \beta_6 Admin_i^2 \\ &+ \beta_7 Parents_i + \beta_8 Counselling_i + \beta_9 CCA_i + \beta_{10} CCA_i^2 + \beta_{11} ProfessionalDevelopment_i \\ &+ \beta_{12} SchoolManagement_i + \beta_{13} OtherTasks_i + \gamma X_i + \varepsilon_i \quad (2) \end{aligned}$$

A statistically significant negative β_4, β_6 and β_{10} indicates a concave association for the respective task, stress rises with hours but at a diminishing rate. The implied turning point for each task is reported as $-\beta_{\text{linear}}/(2\beta_{\text{quadratic}})$ and is interpreted only if it falls within the observed support of the relevant hours variable, as reported in Table 1; turning points outside the observed range represent extrapolation and are flagged accordingly. A linear hypothesis test of $\beta_4 = \beta_6 = \beta_{10}$ assesses whether the degree of concavity differs significantly across the three tasks. Three threats to causal interpretation remain. First, reverse causality: more stressed teachers may perceive and report higher task burdens. Second, unobserved school-level confounding: school-level factors may simultaneously drive both workload and stress. Third, selection on unobserved teacher characteristics, such as conscientiousness or coping ability, may mean that teachers who take on heavier non-teaching workloads differ systematically from those who do not. All three threats remain in the cross-sectional design, and all estimates are accordingly interpreted as conditional associations.

4.3 Robustness Checks

Two robustness checks are estimated. First, a piecewise linear specification replaces the quadratic terms for marking, administrative work, and CCA hours as an alternative functional form test. Knots are selected by RSS minimisation restricted to the 25th-75th percentile of each variable's observed distribution, yielding $\tau_M = 3$ hours for marking, $\tau_A = 2$ hours for admin, and $\tau_C = 1$ hour for CCA, all within observed support. Above-threshold slope coefficients are assessed individually and jointly via Wald test. Second, an ordered probit specification replaces WLS as a check on the continuous approximation. Since T4WLOADT is an OECD-constructed composite scale derived from four Likert items via confirmatory factor analysis, not a single ordinal item, the original response categories cannot be recovered directly. Following standard practice for composite scales used as ordered outcomes, T4WLOADT is cut at sample quartiles to form four ordered stress categories. Both checks use TCHWGT weights and cluster-robust standard errors throughout.

5 Empirical Results

5.1 Model 0: Teaching versus non-teaching hours

Table 2 reports the Model 0 estimates (N = 1,703). Both teaching and non-teaching hours are positively associated with workload stress. Each additional teaching hour is associated with a 0.024

unit increase in workload stress ($SE = 0.005, p < 0.001$), while each additional non-teaching hour is associated with a 0.012 unit increase ($SE = 0.004, p = 0.001$). The teaching hours coefficient is approximately twice the magnitude of the non-teaching hours coefficient. More importantly, the aggregate non-teaching coefficient masks heterogeneity across task types, motivating the task-disaggregated analysis in Models 1 and 2.

Table 2: Survey-weighted regression estimates: workload stress on teaching and non-teaching hours, Singapore *TALIS* 2024

Variable	β	SE
A. Task hours (hours/week)		
Teaching hours (TT4G15)	0.024***	0.005
Total non-teaching hours	0.012**	0.004
B. Controls		
Gender (female=1)	0.259**	0.081
Teacher qualification	-0.158	0.155
Permanent contract	0.442**	0.135
Full-time employment	-0.370.	0.203
Multi-school	-0.011	0.429
Collab. culture	-0.550**	0.165
Peer support	-0.235	0.218
Exp. total: 6–10yr	0.052	0.189
Exp. total: 11–20yr	-0.226	0.153
Exp. total: >20yr	-0.550***	0.156
Exp. school: 6–10yr	-0.160	0.152
Exp. school: 11–20yr	-0.063	0.098
Exp. school: >20yr	0.050	0.201
Intercept	9.698***	0.293

*Significance codes: *** $p < .001$, ** $p < .01$, * $p < .05$, . $p < .10$. Survey-weighted OLS (TCHWGT), cluster-robust SE at school level. $N = 1{,}703$ (three observations dropped due to missing TT4G15). Dependent variable: Workload Stress Scale (T4WLOADT). Base category for experience: ≤ 5 years.*

5.2 Model 1: Linear specification

Table 3 reports the Model 1 estimates. Marking is the only task variable that is large and statistically significant: each additional hour of marking per week is associated with a 0.065 unit increase in workload stress ($SE = 0.011$, $p < 0.001$), equivalent to 0.032 international standard deviations. This is close to the 0.059 standard deviation effect reported by Jerrim and Sims (2021) for marking across five Anglophone systems. Other tasks is also positive and significant ($\beta = 0.030$, $p = 0.001$). Teamwork ($\beta = -0.056$, $p = 0.019$), student counselling ($\beta = -0.039$, $p = 0.022$), and school management ($\beta = -0.053$, $p = 0.014$) are all negatively associated with stress. These nega-

tive associations are most plausibly explained by selection rather than protective effects: teachers assigned to management roles or counselling duties may systematically differ from those who are not, in ways that are correlated with lower stress independently of hours worked. General admin, planning, professional development, parent communication, and CCA are not significant.

Among the controls, teachers on permanent contracts report higher stress ($\beta = 0.308$, $p = 0.046$), while those with a collaborative school culture report lower stress ($\beta = -0.499$, $p = 0.003$). Teachers with more than 20 years of experience report significantly lower stress than those with five or fewer years ($\beta = -0.534$, $p < 0.001$).

5.3 Model 2: Quadratic specification

Model 2 adds quadratic terms for marking, admin, and CCA hours simultaneously. Two findings stand out. First, for marking, the linear term is positive and significant ($\beta = 0.133$, $SE = 0.026$, $p < 0.001$) and the quadratic term is negative and significant ($\beta = -0.004$, $SE = 0.001$, $p < 0.01$). This means stress increases with marking hours, but the rate of increase slows down at higher hours. The implied turning point sits at 18.3 hours per week, above the 95th percentile of the marking distribution (15 hours); because so few teachers report hours near this value, the turning point reflects the algebraic shape of the fitted curve rather than a pattern the data can directly confirm, and within the range teachers report, the marginal stress effect of marking remains positive throughout.

Second, for administrative work, the linear term is on the boundary of conventional significance ($\beta = 0.090$, $SE = 0.046$, $p = 0.050$) and should be interpreted with caution; the quadratic term is more clearly significant ($\beta = -0.004$, $p = 0.047$). For CCA, neither the linear nor the quadratic term is significant ($p = 0.322$ and $p = 0.329$ respectively), suggesting no detectable non-linear association between CCA hours and stress. A linear hypothesis test cannot reject the null that the three quadratic coefficients are equal across tasks ($\chi^2 = 0.004$, df = 2, p = 0.998); given the test's evident lack of power, this is read as inconclusive rather than as confirmation that the three tasks share an identical non-linear shape.

5.4 Interpretation

The main finding is that marking is the task most strongly associated with workload stress among Singapore lower-secondary teachers, consistent with prior evidence. Model 2's quadratic specification shows the marginal stress effect of an extra marking hour gets smaller as hours rise. Raising marking hours by two, from 3h to 5h, predicts a 0.208-unit stress increase. The same two-hour increase higher up the distribution, from 8h to 10h, predicts only 0.135 units. The difference is real but modest, about 1.5 times larger at the low end than at the high end. This could reflect habituation among teachers who routinely carry heavy marking loads, or it could reflect that teachers

new to high marking burdens experience the sharpest stress response. These explanations cannot be distinguished in cross-sectional data. CCA hours show no significant non-linear association with stress once other tasks are held constant, despite being a recognised component of teacher workload in Singapore secondary schools.

The quadratic turning points sit above the 95th percentile of each variable's observed distribution: marking's turning point is 18.3 hours against an observed $p95$ of 15 hours, and administration's is 12.4 hours against 10 hours (Table 1). Because so few teachers report hours near these values, the turning points reflect the algebraic shape of the fitted equation rather than a pattern the data can directly confirm. Within the range of hours that Singapore teachers report, the association between hours and stress is increasing throughout, and the model provides no evidence that it ever flattens or declines at realistic workloads. These turning points are therefore best understood as extrapolations rather than thresholds that any teacher in the sample approaches and should not be read as policy targets.

A different kind of non-linearity shows up in the piecewise model in Section 5.5 below, where the break sits at low hours, not high. Below three hours of marking, each extra hour adds 0.421 points to stress, while above that point the slope drops to 0.031, about a 13.5-fold fall. Admin shows the same shape, with a slope of 0.286 below two hours and 0.022 above it, a 12.7-fold drop close to marking's. So, the steepest rise in stress belongs to teachers with the lightest marking and admin loads, not the heaviest, and this holds for both tasks. It runs against the usual expectation that the most burdened teachers would gain the most from any cut in hours.

This result calls for caution, not immediate policy translation. A steep low-hours slope fits at least three explanations, and the data here cannot rule out any of them. One is a genuine threshold effect: a small amount of marking may carry real start-up costs, such as setting up materials or switching context. These costs ease once a teacher is already in the workflow. Another is a floor or composition effect. Teachers reporting near-zero marking or admin hours may hold different kinds of roles (part-time, early-career, or specialist positions that carry lighter loads by design). If so, the steep low-hours slope may reflect the characteristics of teachers who happen to have light marking loads, rather than what would happen to a typical teacher if their marking were reduced to that level. A further possibility is reverse causality: teachers already under less stress may be the ones assigned, or who select into, lighter marking loads. Telling these apart would need panel data, or some outside source of variation in task assignment, neither of which *TALIS* 2024's cross-sectional design provides.

The joint equality test gives $\chi^2 = 0.004$, df $= 2$, and $p = 0.998$. It cannot distinguish the curvature of marking, admin, and CCA. The test has little power here, though. This makes the result inconclusive, not proof of a shared non-linear shape. It carries little weight to the paper's main argument.

Table 3: Survey-weighted regression estimates: workload stress on non-teaching task hours, Singapore *TALIS* 2024

	Model 1 – Linear		Model 2 – Quadratic	
Variable	β	SE	β	SE
A. Non-teaching task hours (hours/week)				
Planning	0.010	0.012	0.007	0.013
Teamwork	-0.056*	0.023	-0.060*	0.024
Marking	0.065***	0.011	0.133***	0.026
Marking^2			-0.004**	0.001
General admin	0.030	0.021	0.090*	0.046
Admin^2			-0.004*	0.002
Parent communication	-0.002	0.044	-0.019	0.044
Student counselling	-0.039*	0.017	-0.043*	0.017
CCA	0.016	0.021	0.044	0.044
CCA^2			-0.003	0.003
Prof. development	0.012	0.030	0.007	0.030
School management	-0.053*	0.021	-0.055*	0.023
Other tasks	0.030**	0.009	0.027**	0.009
B. Controls				
Gender (female=1)	0.163.	0.084	0.156.	0.082
Teacher qualification	-0.102	0.163	-0.094	0.163
Permanent contract	0.308*	0.153	0.289.	0.151
Full-time employment	-0.170	0.206	-0.176	0.205
Multi-school	0.117	0.415	0.086	0.406
Collab. culture	-0.499**	0.166	-0.501**	0.168
Peer support	-0.230	0.214	-0.205	0.210
Exp. total: 6–10yr	0.078	0.181	0.081	0.181
Exp. total: 11–20yr	-0.191	0.146	-0.183	0.146
Exp. total: >20yr	-0.534***	0.151	-0.508***	0.149
Exp. school: 6–10yr	-0.127	0.139	-0.131	0.138
Exp. school: 11–20yr	-0.068	0.102	-0.063	0.102
Exp. school: >20yr	0.004	0.202	0.018	0.199
Intercept	9.998***	0.308	9.724***	0.308

Implied turning points (Model 2): Marking = 18.3h (above p95); Admin = 12.4h (above p95); CCA = 6.4h (not significant). Joint equality test (quadratic coefs): $\chi^2 = 0.004$, $df = 2$, $p = 0.998$. *Significance codes: *** $p < .001$, ** $p < .01$, * $p < .05$, † $p = .050$ (borderline), . $p < .10$. Survey-weighted OLS (TCHWGT), cluster-robust SE at school level.* $N = 1{,}706$. *Cluster-robust standard errors based on 155 school clusters.*

5.5 Robustness checks

Piecewise specification: The piecewise check uses knots at $\tau_M = 3$ hours for marking, $\tau_A = 2$ hours for admin, and $\tau_C = 1$ hour for CCA, selected by RSS minimisation restricted to the 25th–75th percentile of each variable's distribution. The results confirm the quadratic finding: stress rises steeply below the marking threshold ($\beta = 0.421$, $p < 0.001$) and the above-threshold slope is much flatter (total post-threshold slope = 0.031). The same pattern holds for admin (post-threshold slope = 0.022). The CCA hinge is not significant ($p = 0.098$). The joint Wald test rejects the null that all three hinge coefficients are zero ($F = 21.55$, $p < 0.001$), and the equality test cannot reject the null that the deceleration rates are equal across the three tasks ($\chi^2 = 1.10$, df = 2, $p = 0.578$), suggesting the concave pattern is consistent in magnitude across marking, admin, and CCA.

Table 4: Robustness checks: piecewise linear specification and ordered probit, Singapore TALIS 2024

	RC 1 – Piecewise Linear			RC 2 – Ordered Probit		
Variable	β	SE	Codes	Coef.	SE	Codes
A. Non-teaching task hours						
Planning	—	—		0.009	0.008	
Teamwork	—	—		-0.032	0.015	*
Marking (below $\tau = 3$h)	0.421	0.054	***	0.047	0.008	***
Hinge marking (above $\tau = 3$h)	-0.389	0.059	***	—	—	
Post-threshold slope	0.031					
Admin (below $\tau = 2$h)	0.286	0.112	***	0.009	0.014	
Hinge admin (above $\tau = 2$h)	-0.264	0.109	*	—	—	
Post-threshold slope	0.022					
Parent communication	—	—		0.004	0.030	
Student counselling	—	—		-0.022	0.012	.
CCA (below $\tau = 1$h)	0.286	0.170		0.013	0.012	
Hinge CCA (above $\tau = 1$h)	-0.304	0.182		—	—	
Post-threshold slope	-0.018					
Prof. development	—	—		0.008	0.015	
School management	—	—		-0.025	0.016	
Other tasks	—	—		0.014	0.005	**
B. Model fit / joint tests						
Joint Wald (hinges = 0)	F = 21.55, df = 3,129, $p < .001$					
Equality test (hinges equal)	$\chi^2 = 1.10$, df = 2, $p = .578$					

Knots selected by RSS minimisation restricted to p25–p75 of each variable's distribution: $\tau_M = 3$h, $\tau_A = 2$h, $\tau_C = 1$h. — denotes task variables not interacted with hinge terms (entered linearly; coefficients omitted for brevity). Ordered probit (RC 2) cuts T4WLOADT at sample quartiles into four ordered categories; RC2 reports z-statistics, while RC1 reports t-statistics. RC2 coefficients are reported in the same direction as WLS throughout, so that positive values indicate higher workload stress, consistent with Models 1 and 2. Significance codes: *** $p < .001$, ** $p < .01$, * $p < .05$, . $p < .10$. $N = 1{,}706$.

Ordered probit: The ordered probit produces the same pattern as Model 1. Marking remains the strongest predictor ($z = +6.316$, consistent with the WLS direction given the cumulative probit

sign convention) by a wide margin. All significant coefficients from Model 1 retain their sign and significance. Overall, the ordered probit results support the robustness of the main findings to the choice of estimator.

6 Conclusion

Using TALIS 2024 Singapore data (N = 1,706), this study finds that non-teaching hours vary substantially in their stress associations across task types. Both teaching and non-teaching hours are positively associated with stress. Marking shows the strongest association, with each hour linked to a 0.065-unit rise in stress. Within the hours teachers report, this relationship keeps rising throughout. The increase does slow somewhat at higher hours. The model's implied turning point sits at 18.3 hours, above the 95th percentile of observed marking hours. It is better read as an extrapolation, not a real threshold. A similar pattern holds for admin, with a turning point of 12.4 hours, also outside the observed range. CCA hours show no significant link to stress once other tasks are held constant. These findings speak to the policy debate, with some caution attached. In November 2025, MPs called on MOE to set measurable targets for non-teaching hour reduction. The piecewise results point to larger per-hour gains among teachers with the lightest loads, not the most burdened. The marking gradient falls roughly 13.5-fold above a 3-hour weekly threshold. The admin gradient falls roughly 12.7-fold above a 2-hour threshold. The quadratic results agree, though more modestly. A 5-to-3-hour cut in marking predicts a 0.208-unit stress drop, against 0.135 units for a cut from 10 to 8 hours. Both specifications agree on the direction of the evidence, though the estimates are conditional associations rather than causal effects and should be read as guides for future research.

This evidence cannot pin down a precise safe maximum for marking or admin hours, but it does suggest where policy attention would matter most. Because the steepest stress gradients sit at low hours rather than high ones, marginal relief is likely to help lightly-loaded teachers more than the most burdened. CCA, meanwhile, shows no robust link to stress despite rising hours, making it a weaker candidate for a measurable target. A target grounded in these results would therefore focus on marking and admin at the lighter end of the distribution, rather than a single uniform hours cap. Several limitations apply. The data are cross-sectional, so causal claims cannot be made. The ordered probit check uses quartile-cut stress categories since the original response options are not recoverable. The quadratic specification imposes symmetric curvature, though the piecewise check partially addresses this. Some demographic controls remain unavailable in the Singapore TALIS 2024 public-use file. Subject to these caveats, this study provides the first Singapore-specific, task-level evidence on the non-linear workload-stress association, and a quantitative basis for the measurable targets Parliament has requested.

7 References

References

[1] Angrist, J. D., & Pischke, J.-S. (2009). *Mostly harmless econometrics: An empiricist's companion*. Princeton University Press.

[2] Bakker, A. B., & Demerouti, E. (2007). The job demands-resources model: State of the art. *Journal of Managerial Psychology, 22*(3), 309–328. https://doi.org/10.1108/02683940710733115

[3] Chan, K. B., Lai, G., Ko, Y. C., & Boey, K. W. (2000). Work stress among six professional groups: the Singapore experience. *Social Science and Medicine, 50*(10), 1415–1432. https://doi.org/10.1016/S0277-9536(99)00397-4

[4] Hakanen, J. J., Bakker, A. B., & Schaufeli, W. B. (2006). Burnout and work engagement among teachers. *Journal of School Psychology, 43*(6), 495–513. https://doi.org/10.1016/j.jsp.2005.11.001

[5] Jerrim, J., & Sims, S. (2021). When is high workload bad for teacher wellbeing? Accounting for the non-linear contribution of specific teaching tasks. *Teaching and Teacher Education, 105*, Article 103395. https://doi.org/10.1016/j.tate.2021.103395

[6] Loh, C. E., & Liew, W. M. (2016). Voices from the ground: The emotional labour of English teachers' work. *Teaching and Teacher Education, 55*, 267–278. https://doi.org/10.1016/j.tate.2016.01.016

[7] Ministry of Education, Singapore. (2019). MOE's continuing efforts to improve teachers' welfare. https://www.moe.gov.sg/news/forum-letter-replies/20190701-moes-continuing-efforts-to-improve-teachers-welfare

[8] Ministry of Education, Singapore. (2025a). Parliamentary reply: Addressing teachers' stress levels and supporting their mental well-being. https://www.moe.gov.sg/news/parliamentary-replies/20251104-addressing-teachers-stress-levels-and-supporting-their-mental-well-being

[9] Ministry of Education, Singapore. (2025b). Singapore teachers embrace digital technologies and benefit from strong professional development: OECD TALIS 2024 study. https://www.moe.gov.sg/news/press-releases/20251007-singapore-teachers-embrace-digital-technologies-and-benefit-from-strong-professional-development-oecd-talis-2024-study

[10] Organisation for Economic Co-operation and Development. (2015). How much time do teachers spend on teaching and non-teaching activities? *Education Indicators in Focus, No. 29*. https://doi.org/10.1787/5js64kndz1f3-en

[11] Organisation for Economic Co-operation and Development. (2020). *TALIS 2018 Results (Volume II): Teachers and School Leaders as Valued Professionals*. OECD Publishing, Paris. https://doi.org/10.1787/19cf08df-en

[12] Organisation for Economic Co-operation and Development. (2024). Teaching and Learning International Survey (TALIS) 2024 Teacher Knowledge Survey (TKS): Main Survey Questionnaire (ISCED 2011 Level 2). https://www.oecd.org/content/dam/oecd/en/about/programmes/edu/talis/talis2024questionnaires/TALIS2024TeacherQuestionnaireISCED1-2-3.pdf

[13] Organisation for Economic Co-operation and Development. (2025a). Results from TALIS 2024: The demands of teaching (Chapter 3). https://www.oecd.org/en/publications/results-from-talis-2024_90df6235-en/full-report/the-demands-of-teaching_0e941e2f.html#chapter-d1e11078-37d6897ffa

[14] Organisation for Economic Co-operation and Development. (2025b). Results from TALIS 2024: Thriving in teaching (Chapter 2). https://www.oecd.org/en/publications/results-from-talis-2024_90df6235-en/full-report/thriving-in-teaching_340c1305.html#chapter-d1e6870-a6bf4cb03c

[15] Organisation for Economic Co-operation and Development. (2025c). Technical notes on analyses in this volume. In *Results from TALIS 2024: The state of teaching*. https://www.oecd.org/en/publications/results-from-talis-2024_90df6235-en/full-report/technical-notes-on-analyses-in-this-volume_286d707e.html#annex-d1e31537-3a19d2697b

[16] Organisation for Economic Co-operation and Development. (2025d). Results from TALIS 2024, Country notes: Singapore. https://www.oecd.org/content/dam/oecd/en/publications/reports/2025/10/results-from-talis-2024-country-notes_eafd703e/singapore_f09da5a5/8e88e29a-en.pdf

[17] Organisation for Economic Co-operation and Development. (2025e). Teaching and Learning International Survey (TALIS) 2024 User Guide. https://one.oecd.org/document/EDU/TALIS/GB(2025)9/en/pdf

[18] Organisation for Economic Co-operation and Development. (2025f). *Results from TALIS 2024: The State of Teaching*. OECD Publishing, Paris. https://doi.org/10.1787/90df6235-en

8 Appendix

This appendix presents supplementary figures and tables referenced in the main text. Figure A1 shows the distribution of the workload stress outcome variable. Figure A2 shows the unweighted distributions of all ten non-teaching task-hour variables. Figure A3 presents scatter plots of workload stress against hours for the tasks most strongly correlated with stress. Figure A4 presents the full weighted Pearson correlation heatmap for all 20 analysis variables. Table A2 documents the sample attrition steps. Figure A5 presents the RSS knot search plots for the piecewise linear specification. Figure A1 presents the distribution of the workload stress scale (T4WLOADT) for the analysis sample. The distribution is approximately symmetric and centred just below the international mean of 10, consistent with Singapore's weighted mean of 9.77. The distribution shows a slight left skew, with a small concentration of teachers at the lower end of the scale reporting minimal workload stress.

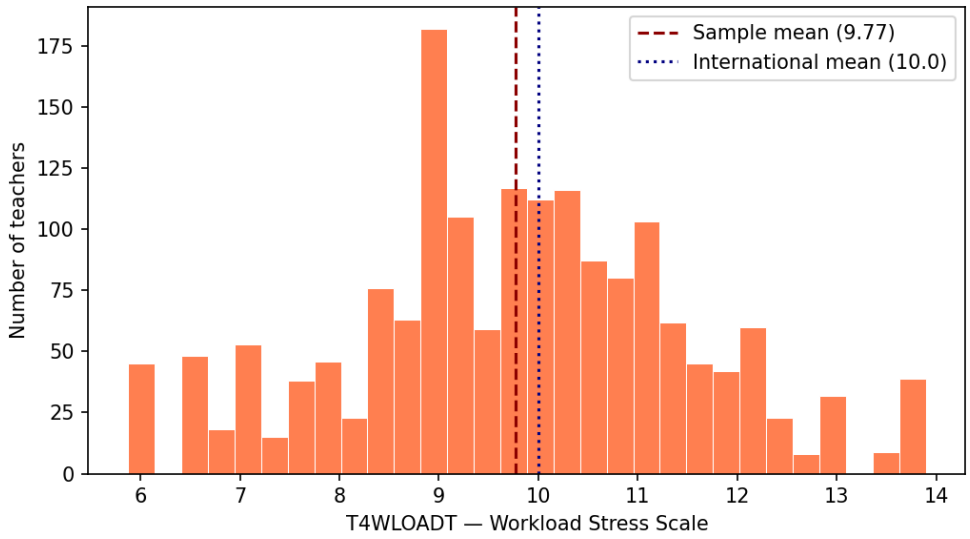


Figure A1. Distribution of workload stress (T4WLOADT), Singapore TALIS 2024.
Notes: Analysis sample (N = 1,706), weighted by TCHWGT. Red dashed line = Singapore weighted mean (9.77). Blue dotted line = international standardised mean (10.0).

Figure A2 presents the unweighted distribution of each of the ten non-teaching task-hour variables for the analysis sample after the 84-hour total exclusion. All task-hour distributions are right-skewed, reflecting the presence of a small number of teachers with very high hours on specific tasks. The green line marks the weighted mean. School management, professional development, parent communication, and CCA show pronounced mass at zero, consistent with the zero percentages reported in Table 1.

Figure A3 presents scatter plots of workload stress (T4WLOADT) against hours for the task variables most strongly correlated with stress in the bivariate analysis: marking (r = 0.19), other tasks (r = 0.13), and planning (r = 0.10). Each scatter plot is survey-weighted and includes a linear fit

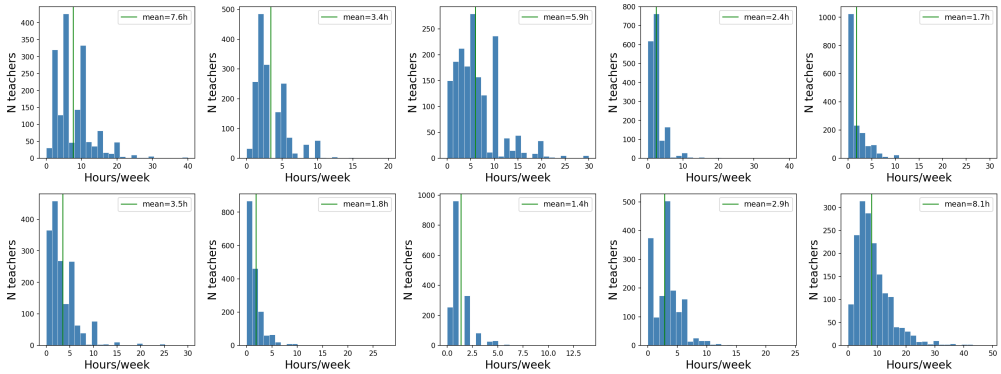


Figure A2. Non-teaching task-hour distributions, Singapore TALIS 2024.
Notes: Analysis sample (N = 1,706). Each panel shows the unweighted distribution after 84-hour total exclusion. Green line marks the weighted mean.

line. The positive slopes confirm the direction of the bivariate correlations reported in Section 3.4. The substantial dispersion around each fit line illustrates why the conditional regression estimates in Section 4 are needed to isolate the independent contribution of each task holding the others constant.

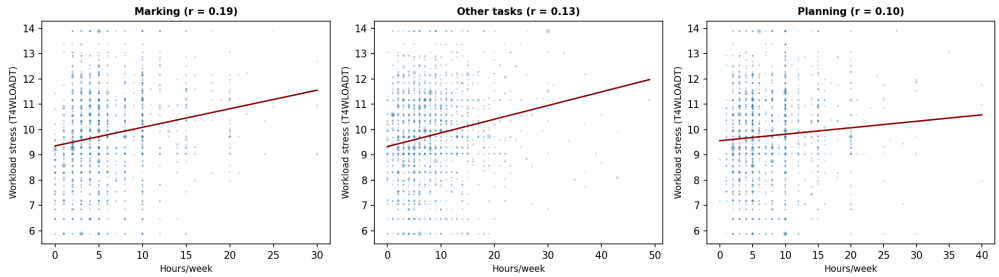


Figure A3. Survey-weighted scatter plots of workload stress against hours for the most strongly correlated task variables, Singapore TALIS 2024.
Notes: Analysis sample (N = 1,706), weighted by TCHWGT. Variables shown: marking (r = 0.19), other tasks (r = 0.13), planning (r = 0.10). Linear fit line included.

First, no pair of task-hour variables exceeds $r = 0.34$ (marking and planning), confirming that multicollinearity is not a concern in either regression specification. Second, the correlations between task variables are mostly positive and moderate, consistent with the hypothesis that teachers who spend more time on one non-teaching task also tend to spend more time on others. Third, the correlations between each task variable and workload stress range from $r = -0.13$ for school management to $r = 0.19$ for marking, with marking showing the strongest positive association. The school management correlation is negative, consistent with the selection interpretation in Section 5.2: teachers assigned to management duties may differ systematically from those who are not, in ways correlated with lower stress independently of hours.

Table A2 documents the sample construction steps from the full Singapore TALIS 2024 dataset to the analysis sample. Starting from 3,355 lower-secondary teachers, approximately 1,126 did

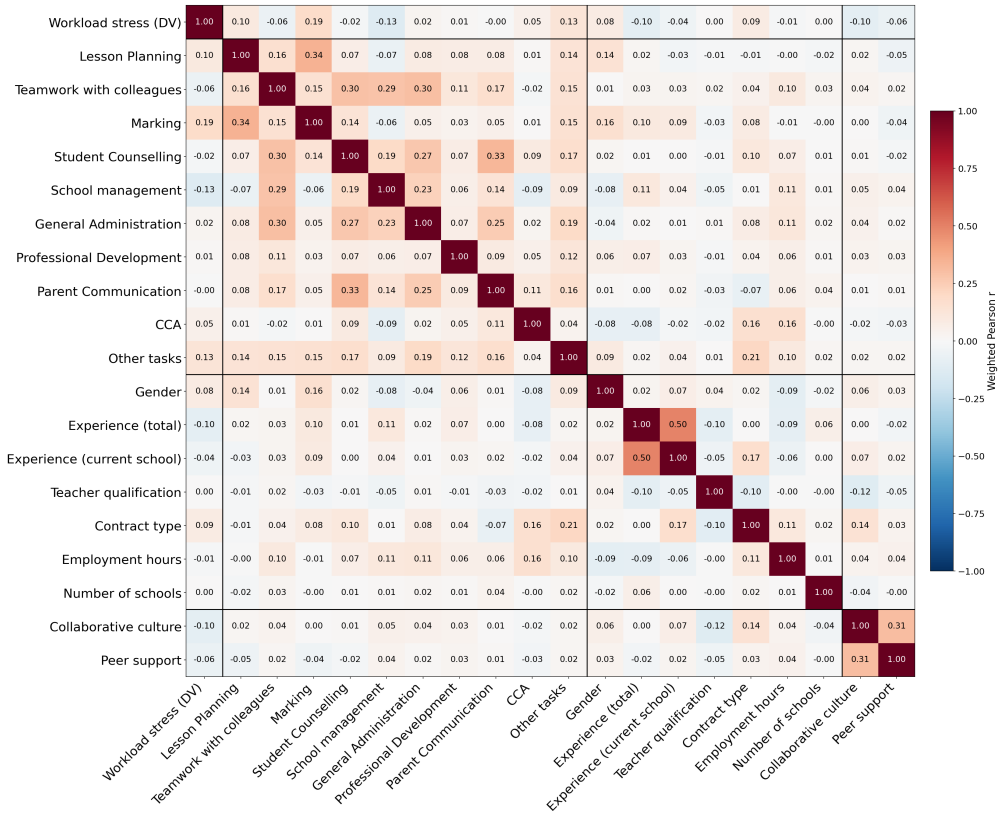


Figure A4. Weighted Pearson correlation heatmap — all analysis variables, Singapore TALIS 2024 (N = 1,706).

Notes: Weighted by TCHWGT. Colour scale: dark red = strong positive ($r = 1.0$), dark blue = strong negative ($r = -1.0$). Variable groups: DV (workload stress), IVs (10 task-hour variables), Controls, and Resources.

not receive the task-hour and outcome questionnaire block, missing by design due to random questionnaire rotation. Listwise deletion across all analysis variables, including within-form item non-response, retains 1,768 teachers with complete data. The 84-hour exclusion then removes 62 teachers (3.5%) reporting implausibly high total weekly non-teaching hours, yielding the final analysis sample of 1,706 teachers.

Figure A5 presents the RSS curves from the knot search underlying the piecewise linear specification. For marking, the RSS curve shows a sharp, well-defined minimum, with the unrestricted global minimum at $\tau = 2$ hours and the IQR-restricted minimum at $\tau = 3$ hours; the two are close and the curve is steep in this region, confirming that the structural break is genuine rather than noise-driven. For general administration, the unrestricted minimum falls at $\tau = 1$ hour, at the edge of the observed distribution where observations are sparse; the IQR-restricted minimum of $\tau = 2$ hours is retained on this basis. For CCA, both searches converge on $\tau = 1$ hour, indicating the knot choice is unambiguous. Across all three variables, the RSS curves rise monotonically beyond the minimum, confirming that no competing breakpoint exists at higher hours.

Table A2. Sample construction and exclusions, Singapore TALIS 2024

Step	Dropped	Remaining	Note
Full Singapore ISCED 2 sample	—	3,355	All teachers in 145 public + 10 private schools
Listwise deletion on baseline controls	-1,587	1,768	Rotation form not administered (∼1,126) plus within-form item non-response (461)
84-hour exclusion (>84h total non-teaching)	-62	1,706	Implausibly high reporters excluded; 3.5% of pre-exclusion sample
Primary analysis sample	—	1,706	Used for all descriptive statistics, correlations, and regressions

Notes: Listwise deletion was applied on all analysis variables: T4WLOADT (outcome), all ten task-hour variables (TT4G16A–TT4G16J), employment status (T4TEMPWH), gender (TT4G01), contract type (TT4G09), teacher qualification type (TT4G04), experience (T4TYEXPTT, T4TYEXPTS), number of schools (T4TNSCH), collaborative culture (TT4G64D), and peer support (TT4G64H).

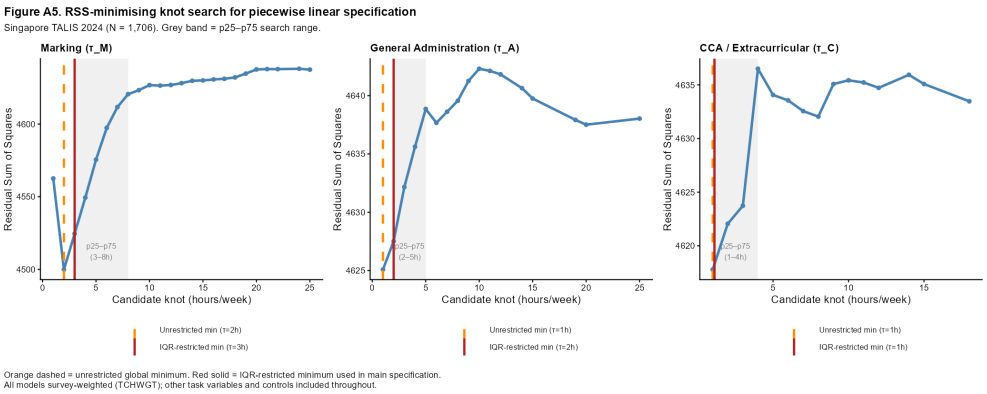


Figure A5. RSS-minimising knot search for piecewise linear specification. Singapore TALIS 2024 (N = 1,706).

Notes: Y-axis = RSS at each candidate knot (x-axis, hours/week); lower = better fit. Grey band = p25–p75 search range. Orange dashed = unrestricted global minimum; red solid = IQR-restricted minimum retained in the main specification. All models survey-weighted (TCHWGT); other task variables and controls held constant throughout.